Case Study Practical

1) A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Step	**What to Check**
1. Check the power adapter and cable	Ensure the charger is connected properly, not damaged, and the power outlet is working.
2. Check the charging indicator light	Look for the charging LED. If it doesn't light up, the charger or charging port may be faulty.
3. Perform a power reset	Disconnect the charger, remove the battery (if removable), hold the power button for 20–30 seconds, then reconnect power. This removes residual electrical charge.
4. Inspect the battery	Check whether the battery is swollen or damaged. Try turning on the laptop with only the charger connected.
5. Test hardware components	Listen for fan noise, check RAM, or connect another charger. This helps identify motherboard, RAM, or power supply problems.

2) You receive a helpdesk ticket: "My monitor is showing a blank screen, but the CPU seems to be running." Write a step-by-step troubleshooting flow to identify and resolve the issue.

Step 1: Check the monitor power

- Ensure the monitor is turned on.
- Verify the power cable is connected.
- Check if the power LED is lit.

Step 2: Check display cables

- Ensure the HDMI, DisplayPort, VGA, or DVI cable is securely connected.
- Try another cable if available.

Step 3: Check the correct input source

- Use the monitor menu to select the correct input (HDMI, VGA, DP, etc.).

Step 4: Restart the computer

- Restart the PC and observe whether the display appears.

Step 5: Test with another monitor

- Connect another monitor to determine whether the issue is with the monitor or the computer.

Step 6: Check the graphics card

- Ensure the graphics card is seated properly.
- Connect the monitor to another display port if available.

Step 7: Check RAM

- Turn off the PC, remove and reseat the RAM modules.

Step 8: Update or reinstall display drivers

- If the display works in Safe Mode, reinstall or update the graphics driver.

3) A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Step 1: Check the mouse battery
- Replace or recharge the batteries.

Step 2: Turn the mouse off and on
- Ensure the power switch is ON.

Step 3: Check the USB receiver
- Remove and reconnect the USB receiver.

Step 4: Try another USB port
- Connect the receiver to a different USB port.
- If it works, the original USB port is faulty.

Step 5: Test the mouse on another computer
- If it still doesn't work, the mouse is likely defective.

Step 6: Check Bluetooth (if applicable)
- Make sure Bluetooth is enabled and the mouse is paired.

Step 7: Check Device Manager
- Look for driver errors.
- Update or reinstall the mouse driver if necessary.

Step 8: Verify system settings
- Ensure the mouse is enabled in Windows settings.
- Restart the computer if required.

4) Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hardware Cause	**How to Check**
1. Dust buildup in fans and heatsink	Open the case and inspect for dust. Clean the fans, heatsink, and air vents using compressed air.
2. Faulty cooling fan	Check whether the CPU and case fans are spinning properly. Replace any fan that is slow or not working.
3. Dried or damaged thermal paste	Remove the CPU cooler and inspect the thermal paste. Clean off the old paste and apply a fresh layer if it has dried out.

Additional Checks

- Monitor CPU and GPU temperatures using software like HWMonitor or Core Temp.
- Ensure proper airflow inside the PC.
- Verify that all fans are connected correctly.